MIZORAM PUBLIC SERVICE COMMISSION

GENERAL COMPETITIVE EXAMINATION FOR RECRUITMENT TO THE POST OF JUNIOR GRADE OF MIZORAM PLANNING, ECONOMIC & STATISTICAL SERVICE UNDER PLANNING & PROGRAMME IMPLEMENTATION DEPARTMENT. (DECEMBER, 2025)

MATHEMATICS-I

Time Allowed : 3 hours | Full Marks : 100

Marks for each question is indicated against it.
All statistical notations carry their standard meanings.
Use of a scientific calculator is permitted.

Directions (Question No. 1 – 3): Attempt any 2 (two) questions.

1. (a) Define continuity of a function of single variable at a point. Discuss the continuity of the function

$$f(x)=\begin{cases}\dfrac{sin2x}{x}, & x\neq 0\\ 1, & x=0\end{cases}$$ at the origin. **(5)**

(b) Let $f:D\to\mathbb{R}$ where $D\subseteq\mathbb{R}^2$ and let (a,b) be a point in D. If f_x exists throughout a neighbourhood of (a,b) and f_y exists at (a,b), then show that for any point $(a+h,b+k)$ of this neighbourhood,

$$f(a+h,b+k)-f(a,b)=hf_x(a+\theta h,b+k)+k\left[f_y(a,b)+\eta\right],$$

where $0<\theta<1$ and η is a function of k tending to zero with k. **(5)**

2. Show that the sequence $\{f_n\}$, where $f_n=\dfrac{nx}{1+n^2x^2}$ is pointwise convergent but not uniformly convergent at every point on the real line. **(10)**

3. (a) Find the directional derivative of $f(x,y)=x^3+ysinx$ at $\left(0,\dfrac{\pi}{4}\right)$ in the direction of $\vec{v}=3\hat{i}+4\hat{j}$. **(5)**

(b) Using Taylor's theorem, show that the expansion of $\sin(xy)$ in powers of $(x-1)$ and $\left(y-\dfrac{\pi}{2}\right)$ upto and including the second degree term is

$$1-\frac{\pi^2}{8}(x-1)^2-\frac{\pi}{2}(x-1)\left(y-\frac{\pi}{2}\right)-\frac{1}{2}\left(y-\frac{\pi}{2}\right)^2$$ **(5)**

Directions (Question No. 4 – 6): Attempt any 2 (two) questions.

4. Let $(X,\|\cdot\|_1)$ and $(X,\|\cdot\|_2)$ be two normed linear spaces on the same vector space X. Then

(a) Show by explicit calculation that the norms

$$\|x\|_1 = |x_1| + |x_2|, \qquad \|x\|_2 = \sqrt{x_1^2 + x_2^2}$$

are equivalent on $\mathbb{R}^2$. **(6)**

(b) Show that the corresponding metric topologies are identical. **(4)**

5. Let $T : l_\infty \to \mathbb{R}$ be defined by

$$T(x_1, x_2, x_3, \cdots) = \lim_{n\to\infty} x_n,$$

for every convergent bounded sequence (x_n).

(a) Prove that T is a bounded linear functional and find its norm. **(6)**

(b) Use the Hahn-Banach theorem to extend T to all of l_∞ (no need to construct it explicitly). **(4)**

6. Let $H = L^2[0,1]$ be a Hilbert space with the inner product

$$\langle f, g\rangle = \int_0^1 f(x)\overline{g(x)}\,dx.$$

(a) Verify the parallelogram law and use it to define a norm induced by the inner product. **(4)**

(b) Let $T : H \to H$ be defined by $(Tf)(x) = x f(x)$. Then, show that T is self-adjoined and bounded. **(6)**

Directions (Question No. 7 – 9): Attempt any 2 (two) questions.

7. Define a group. Prove that the order of each subgroup of a finite group is a divisor of the order of the group but the converse need not be true. **(2+8=10)**

8. Define a Euclidean ring. Show that every Euclidean ring is a principal ideal domain. **(2+8=10)**

9. (a) Show that every field is an integral domain. **(5)**

(b) Let K / F be a finite extension of degree n and L / K be a finite extension of degree m, then prove that L / F is a finite extension of degree mn, that is

$$[L : F] = [L : K][K : F].$$ **(5)**

Directions (Question No. 10– 12): Attempt any 2 (two) questions.

10. (a) Let $A = [0,1) \cup (1,2] \subset \mathbb{R}$ with the standard topology.

(i) Find the closure, interior and boundary of A. **(3)**

(ii) Decide whether A is open, closed, or neither, with reasons. **(2)**

(b) Let $f:(X,\tau_X)\to(Y,\tau_Y)$ be a mapping between two topological spaces. Prove that a function f is continuous if and only if for every $A\subseteq X$, $f(\overline{A})\subseteq\overline{f(A)}$, where '$\overline{A}$' denotes the closure of A. **(5)**

11. Let $\mathbb{R}$ carry the lower limit topology whose basis consists of half-open intervals $[a,b)$.

(i) Show that this space is Hausdorff. **(3)**

(ii) Determine whether it is first countable, and/or second countable. **(4)**

(iii) Compare it to the usual Euclidean topology on $\mathbb{R}$: Is it finer, coarser, or incomparable? **(3)**

12. Let $X=[0,1]\times[0,1]\subset\mathbb{R}^2$ with the product (Euclidean) topology and let $Y=X-\left\{\left(\frac{1}{2},\frac{1}{2}\right)\right\}$.

(i) Prove that X is compact. **(2)**

(ii) Show that X is path-connected and connected. **(2)**

(iii) Decide whether Y is compact, connected, and/or path-connected. Justify. **(6)**

Directions (Question No. 13 – 15): Attempt any 2 (two) questions.

13. Derive the equation of motion for a simple pendulum of length l and mass of bob m by using Lagrange's equations. **(10)**

14. Using calculus of variations, show that the shortest curve between two points on a plane is a straight line. **(10)**

15. Show that the transformation $Q=\log\left(\frac{1}{q}\sin p\right)$, $P=q\cot p$ is canonical. **(10)**

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